

The Financial Economics and Spectral Geometry of Promissory Money

Cash-to-Coin Ratios as Monetary and Financial Shocks

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Abstract

This paper develops a general theory of promissory money backed, wholly or partially, by coinage. The central state variable is the cash-to-coin ratio

$$\chi_t = \frac{C_t}{M_t},$$

where C_t denotes the outstanding nominal stock of promissory notes and M_t denotes a specified coin base. The principal theoretical innovation is to treat unexpected variation in χ_t , rather than merely variation in the money stock, as a monetary and financial shock.

The theory extends a preceding spectral-geometric theory of coinage. Coin geometry, material density, thickness, perforation, constant-width structure, denomination, production technology, and installed physical stocks jointly influence the quantity and characteristics of the monetary base. Promissory notes constitute a second layer of monetary architecture. The interaction of the physical base and the promissory layer generates liquidity amplification, redemption constraints, confidence feedback, financial phase transitions, and macroeconomic transmission.

The framework integrates monetary economics, finance, geometry, mechanics, materials science, chemistry, behavioral science, psychology, psychiatry, philosophy, political economy, international relations, welfare economics, statistics, computer science, algorithms, data science, artificial intelligence, and systemic-risk analysis. The resulting system treats money as a coupled physical, contractual, informational, and strategic object.

The paper ends with “The End”

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1 Introduction

Money may be represented physically, contractually, or electronically. Coinage represents a particularly important physical form of money because its production, circulation, durability, geometry, material composition, and recognizability are all subject to physical constraints. Promissory notes add a second layer: a transferable claim whose monetary usefulness depends not only upon its physical production but also upon redemption, institutional credibility, legal enforceability, and collective belief.

This paper develops a theory in which promissory money is related explicitly to an underlying coin base. Let

$$C_t \geq 0$$

denote the outstanding nominal value of promissory notes and let

$$M_t > 0$$

denote the relevant value of coinage supporting the system.

Define the cash-to-coin ratio by

$$\chi_t = \frac{C_t}{M_t}. \quad (1)$$

The central proposition is that χ_t is not merely an accounting ratio. Its innovations can constitute monetary shocks.

Let

$$x_t = \log \chi_t.$$

Then define

$$\varepsilon_{\chi,t} = x_t - \mathbb{E}_{t-1}[x_t]. \quad (2)$$

Thus,

$$\boxed{\varepsilon_{\chi,t} = \log \left(\frac{C_t}{M_t} \right) - \mathbb{E}_{t-1} \left[\log \left(\frac{C_t}{M_t} \right) \right]}. \quad (3)$$

The objective of the paper is to construct a complete theory around the causal architecture

$$\boxed{\begin{array}{ccccccc} \text{Coin} & \rightarrow & \text{Coin} & \rightarrow & \text{Promissory Notes} & \rightarrow & \text{Monetary} & \rightarrow & \text{Macro-Financial} \\ \text{Geometry} & & \text{Base} & & \text{Coinage} & & \text{Shock} & & \text{Dynamics} \end{array}}.$$

The analysis proceeds through five stages:

1. single-instrument mechanics;
2. static financial phase structure;
3. multi-denomination promissory portfolios;
4. dynamic installed-stock and redemption dynamics;
5. empirical identification and econometric analysis.

The resulting theory treats physical money and promissory money as a coupled dynamical system.

2 Conceptual Foundations

2.1 Money as a Physical and Contractual System

Let the total transaction medium be

$$\mathcal{L}_t = M_t + C_t. \quad (4)$$

Using $C_t = \chi_t M_t$ gives

$$\mathcal{L}_t = (1 + \chi_t) M_t. \quad (5)$$

The ratio χ_t therefore determines the degree to which the nominal transaction system exceeds the coin base.

Define the monetary layering multiplier

$$\mu_t = 1 + \chi_t. \quad (6)$$

The multiplier is not assumed constant. It is an endogenous state variable.

This distinction is fundamental. In conventional monetary analysis, the quantity of money may itself be the principal object of interest. Here, the structural relationship between a promissory layer and a physical base is elevated to equal importance.

2.2 The Three Meanings of Coin Backing

The denominator M_t must be defined carefully. At least three concepts are possible.

Let $N_{d,t}$ denote the number of coins of denomination d and F_d their face value. Then nominal backing is

$$M_t^N = \sum_{d=1}^D F_d N_{d,t}. \quad (7)$$

Let $\mathcal{M}_{d,t}$ denote physical mass. Then physical backing is

$$M_t^P = \sum_{d=1}^D \mathcal{M}_{d,t}. \quad (8)$$

Let $V_{d,t}^{\text{coin}}$ denote the market value of the relevant coin reserve. Then market-value backing is

$$M_t^V = \sum_{d=1}^D V_{d,t}^{\text{coin}}. \quad (9)$$

Therefore,

$$\mathbf{x}_t = \begin{pmatrix} \chi_t^N \\ \chi_t^P \\ \chi_t^V \end{pmatrix} = \begin{pmatrix} C_t/M_t^N \\ C_t/M_t^P \\ C_t/M_t^V \end{pmatrix}. \quad (10)$$

These ratios measure different forms of monetary exposure.

Table 1: Alternative cash-to-coin ratios

Ratio	Denominator	Interpretation
χ^N	Nominal coin value	Legal or face-value backing
χ^P	Physical coin mass	Material-resource exposure
χ^V	Market coin value	Valuation and solvency exposure

3 Stage I: Single-Instrument Mechanics

3.1 Promissory Note Production

Consider a promissory note of face value F . Let its production cost be

$$c_C(F, t) = f_C(F) + v_C(F, t), \quad (11)$$

where $f_C(F)$ denotes fixed costs and $v_C(F, t)$ variable costs.

Variable costs may depend upon:

1. substrate;
2. printing technology;
3. security features;
4. inks and pigments;
5. serial numbering;
6. anti-counterfeiting technology;
7. labor;
8. energy;
9. logistics;
10. replacement and destruction.

Thus,

$$v_C(F, t) = v_C(F, p_t^{\text{paper}}, p_t^{\text{polymer}}, p_t^{\text{energy}}, w_t, q_t^{\text{security}}). \quad (12)$$

The production of a note is therefore simultaneously an engineering, economic, and security problem.

3.2 The Static Layering Problem

Let $B(\chi)$ denote transaction and liquidity benefits. Let $\Phi(\chi)$ denote expected financial fragility. Let $K(\chi)$ denote issuance and administrative costs.

The issuing authority solves

$$\max_{\chi \geq 0} \{B(\chi) - \Phi(\chi) - K(\chi)\}. \quad (13)$$

Assume

$$B'(\chi) > 0, \quad (14)$$

$$B''(\chi) < 0, \quad (15)$$

$$\Phi'(\chi) > 0, \quad (16)$$

$$\Phi''(\chi) > 0. \quad (17)$$

For an interior solution,

$$\boxed{B'(\chi^*) = \Phi'(\chi^*) + K'(\chi^*)}. \quad (18)$$

The optimal monetary architecture balances liquidity creation against financial fragility.

Proposition 1. *Suppose $B''(\chi) < 0$, $\Phi''(\chi) > 0$, and $K''(\chi) \geq 0$. Then any interior optimum χ^* is unique.*

Proof. The second derivative of the objective is

$$B''(\chi) - \Phi''(\chi) - K''(\chi) < 0. \quad (19)$$

Therefore the objective is strictly concave. A stationary point, if it exists, is unique and globally optimal. $\square$

4 Liquidity Amplification

From

$$\mathcal{L} = (1 + \chi)M,$$

holding M constant gives

$$\frac{\partial \log \mathcal{L}}{\partial \log \chi} = \frac{\chi}{1 + \chi}. \quad (20)$$

Define

$$\eta_\chi = \frac{\partial \log \mathcal{L}}{\partial \log \chi}. \quad (21)$$

Therefore,

$$\boxed{\eta_\chi = \frac{\chi}{1 + \chi}}. \quad (22)$$

Moreover,

$$\frac{\partial \eta_\chi}{\partial \chi} = \frac{1}{(1 + \chi)^2} > 0. \quad (23)$$

Proposition 2 (Monetary Layering Amplification). *The proportional sensitivity of total transaction balances to proportional changes in the cash-to-coin ratio increases monotonically in χ .*

Proof. Immediate from

$$\frac{\partial \eta_\chi}{\partial \chi} > 0.$$

$\square$

Thus highly layered systems are intrinsically more sensitive to innovations in the ratio.

5 Stage II: Financial Phase Structure

Define four illustrative financial regimes:

$$B_0 : \chi = 0, \quad (24)$$

$$B_1 : 0 < \chi \leq 1, \quad (25)$$

$$B_2 : 1 < \chi \leq \bar{\chi}, \quad (26)$$

$$B_3 : \chi > \bar{\chi}. \quad (27)$$

These correspond respectively to:

1. pure coinage;
2. conservative promissory layering;
3. fractional backing;
4. a potentially fragile regime.

The threshold $\bar{\chi}$ is endogenous:

$$\bar{\chi}_t = \Psi(r_t, \sigma_t, \Gamma_t, \mathcal{R}_t, \mathcal{K}_t), \quad (28)$$

where Γ_t denotes confidence, $\mathcal{R}_t$ redemption capacity, and $\mathcal{K}_t$ institutional credibility.

Therefore,

A cash-to-coin ratio has no universal stability interpretation.

(29)

Its significance depends upon the institutional environment.

5.1 A Phase Diagram

A simple phase representation may be constructed in (r, χ) space.

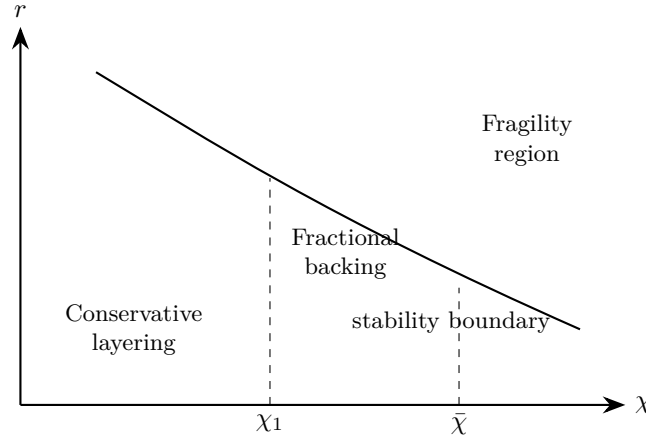


Figure 1: A simple phase representation.

The figure is schematic rather than a universal empirical law.

6 Stage III: Portfolios of Promissory Claims

Suppose the monetary system contains D note denominations:

$$F_1 < F_2 < \dots < F_D.$$

Let

$$C_t = \sum_{d=1}^D C_{d,t}.$$

Define denomination-specific ratios

$$\chi_{d,t} = \frac{C_{d,t}}{M_{d,t}}. \quad (30)$$

Collect them into

$$\boldsymbol{\chi}_t = (\chi_{1,t}, \dots, \chi_{D,t})'. \quad (31)$$

A system-wide objective may be

$$\begin{aligned} \mathcal{J}_t = & \sum_{d=1}^D \omega_d B_d(\chi_{d,t}) - \sum_{d=1}^D \omega_d \Phi_d(\chi_{d,t}) \\ & - \sum_{i < j} L_{ij}^{\text{conf}} P_{ij}^{\text{conf}} - \mathcal{C}_t. \end{aligned} \quad (32)$$

The final term involving L_{ij}^{conf} permits the model to incorporate confusion, substitution, security, or operational losses between denominations.

A shock can now be denomination-specific:

$$\boldsymbol{\varepsilon}_{\chi,t} = \boldsymbol{\chi}_t - \mathbb{E}_{t-1}[\boldsymbol{\chi}_t], \quad (33)$$

where

$$\boldsymbol{\chi}_t = (\log \chi_{1,t}, \dots, \log \chi_{D,t})'. \quad (34)$$

Consequently, two systems with identical aggregate χ_t may exhibit different financial properties if their denomination structures differ.

7 Stage IV: Dynamic Installed Stocks

7.1 Coin Dynamics

Let the coin base evolve according to

$$M_{t+1} = (1 - \delta_t^M)M_t + Q_t^M - W_t^M, \quad (35)$$

where δ_t^M is attrition, Q_t^M new coin production, and W_t^M withdrawal.

7.2 Promissory-Note Dynamics

Let

$$C_{t+1} = (1 - \delta_t^C)C_t + I_t - R_t, \quad (36)$$

where δ_t^C is note attrition, I_t issuance, and R_t redemption or retirement.

Therefore,

$$\chi_{t+1} = \frac{(1 - \delta_t^C)C_t + I_t - R_t}{(1 - \delta_t^M)M_t + Q_t^M - W_t^M}. \quad (37)$$

Taking logarithmic approximations,

$$\Delta \log \chi_t \approx \Delta \log C_t - \Delta \log M_t. \quad (38)$$

Hence the cash-to-coin shock can be decomposed as

$$\boldsymbol{\varepsilon}_{\chi,t} \approx \boldsymbol{\varepsilon}_{C,t} - \boldsymbol{\varepsilon}_{M,t}. \quad (39)$$

This distinction is essential. A rise in χ_t may result from an expansion of promissory money, a contraction of coinage, or both.

8 Redemption Mechanics

Let $\delta_t \in [0, 1]$ denote the fraction of outstanding notes presented for immediate redemption.

Redemption demand is

$$D_t^R = \delta_t C_t. \quad (40)$$

Immediate redemption against the coin base requires

$$\delta_t C_t \leq M_t. \quad (41)$$

Since $C_t = \chi_t M_t$,

$$\delta_t \chi_t \leq 1. \quad (42)$$

Therefore,

$$\boxed{\delta_t^{\max} = \frac{1}{\chi_t}}. \quad (43)$$

Theorem 1 (Cash-to-Coin Amplification and Redemption Fragility). *Suppose*

$$\mathcal{L}_t = (1 + \chi_t) M_t,$$

with $M_t > 0$ and $\chi_t > 0$. Then an increase in χ_t :

1. increases the proportional sensitivity of transaction balances to ratio shocks;
2. increases the quantity of promissory claims per unit of coin backing;
3. decreases the maximum fraction of simultaneous immediate redemption that can be satisfied by the coin base.

Proof. The liquidity elasticity is

$$\eta_\chi = \frac{\chi}{1 + \chi},$$

and

$$\frac{\partial \eta_\chi}{\partial \chi} = \frac{1}{(1 + \chi)^2} > 0.$$

The number of claims per unit of backing rises directly with χ . Finally,

$$\delta^{\max} = \frac{1}{\chi},$$

and

$$\frac{\partial \delta^{\max}}{\partial \chi} = -\frac{1}{\chi^2} < 0.$$

Therefore all three statements hold. □

9 Confidence, Psychology, and Collective Expectations

Money functions partly because economic agents expect other agents to accept it. Promissory money adds an additional expectation:

acceptability + belief in redemption.

Let Γ_t denote confidence. A stylized transition equation is

$$\Gamma_{t+1} = \rho_\Gamma \Gamma_t - a_\chi \chi_t - a_\delta \delta_t + a_K \mathcal{K}_t + \varepsilon_{\Gamma,t+1}. \quad (44)$$

Suppose redemption pressure follows

$$\delta_t = \Lambda(\alpha_\delta - \beta_\Gamma \Gamma_t + \beta_\chi \chi_t + \varepsilon_{\delta,t}), \quad (45)$$

where

$$\Lambda(x) = \frac{1}{1 + e^{-x}}. \quad (46)$$

The system may therefore contain the feedback mechanism

$$\boxed{\chi_t \uparrow \longrightarrow \Gamma_t \downarrow \longrightarrow \delta_t \uparrow \longrightarrow \text{financial stress} \uparrow .}$$

Psychologically, such dynamics involve trust, imitation, salience, availability, loss aversion, and strategic anticipation. Psychiatry may be relevant at the population level only indirectly through stress, panic, impaired judgment, and behavioral responses under extreme crisis; the model does not interpret aggregate monetary behavior as a psychiatric diagnosis.

10 Interest Rates, Including Negative Rates

Let the desired cash-to-coin ratio satisfy

$$\chi_t^* = \chi_0 + \beta_r r_t + \beta_\sigma \sigma_t + \beta_\Gamma \Gamma_t + \varepsilon_{\chi,t}. \quad (47)$$

The sign of β_r is not assumed universal.

If $r_t < 0$, agents may alter their relative demand for physical money, promissory notes, deposits, and other stores of value. The response depends upon institutional design, storage costs, convertibility, transaction technology, denomination structure, and legal restrictions.

Therefore,

$$\boxed{r_t < 0 \not\Rightarrow \Delta\chi_t \text{ has a universally predetermined sign.}} \quad (48)$$

Negative rates are consequently a source of regime dependence rather than a trivial extension of positive-rate analysis.

11 Coupling Promissory Money to Spectral Coin Geometry

Let the outer width of a coin be W and let $w < W$ denote the width of an inner perforation.

Let the material density be

$$\rho(\omega), \quad \omega \in [0, W],$$

with

$$\rho(\omega) = 0, \quad 0 \leq \omega \leq w, \quad (49)$$

$$\rho(\omega) > 0, \quad w < \omega \leq W. \quad (50)$$

Let the constant-width boundary be represented spectrally by

$$h(\theta) = \frac{W}{2} + \sum_{n \in \mathcal{A}} z_n \cos((2n+1)\theta + \phi_n). \quad (51)$$

The spectral coefficients z_n and phases ϕ_n determine deformations compatible with the chosen geometric representation.

Let $\lambda = w/W$ denote normalized perforation. Define a dimensionless mass or material factor

$$m(\mathbf{z}, \lambda) = 1 - \lambda^2 - \sum_n \alpha_n z_n^2. \quad (52)$$

Then, schematically, the physical mass of denomination d may be written

$$\mathcal{M}_{d,t} = N_{d,t} \rho_{d,t} \tau_{d,t} \pi R_{d,t}^2 m(\mathbf{z}_{d,t}, \lambda_{d,t}). \quad (53)$$

Therefore,

$$\Delta \log M_t^P = \sum_d \omega_{d,t}^M [\Delta \log N_{d,t} + \Delta \log \rho_{d,t} + \Delta \log \tau_{d,t} + 2\Delta \log R_{d,t} + \Delta \log m_{d,t}]. \quad (54)$$

Moreover,

$$dm = -2\lambda d\lambda - 2 \sum_n \alpha_n z_n dz_n. \quad (55)$$

Hence,

$$d \log m = \frac{-2\lambda d\lambda - 2 \sum_n \alpha_n z_n dz_n}{m}. \quad (56)$$

The geometry of coinage can therefore enter the promissory-money system through the denominator of χ_t .

The full transmission mechanism is

$$(z_t, \lambda_t) \longrightarrow m_t \longrightarrow M_t \longrightarrow \chi_t \longrightarrow \varepsilon_{\chi,t} \longrightarrow \text{economic and financial outcomes.}$$

12 Physics and Materials Science

The coin is a mechanical object subject to wear, impact, friction, corrosion, deformation, and material fatigue. Its installed stock evolves partly because individual physical objects deteriorate.

A stylized wear equation is

$$\frac{dM_t}{dt} = Q_t - W_t - \delta_M(t)M_t. \quad (57)$$

The attrition rate may depend on

$$\delta_M(t) = \delta_M(T_t, H_t, E_t, \mathcal{U}_t), \quad (58)$$

where T_t denotes temperature exposure, H_t humidity or chemical environment, E_t mechanical exposure, and $\mathcal{U}_t$ circulation intensity.

Chemically, oxidation and corrosion can alter mass, surface properties, and machine recognizability. Engineering design therefore affects not only manufacturing cost but potentially the long-run dynamics of the monetary base.

13 Biology and Human Interaction

Biological considerations enter through human use. Currency is handled, transported, stored, contaminated, cleaned, visually perceived, and distinguished by human sensory and motor systems.

For physical notes, durability depends partly on substrate resistance to folding, moisture, abrasion, microbial degradation, and repeated handling. For coins, tactile differentiation and motor recognition can affect accessibility.

A generalized utility function may include usability:

$$U = U(\text{purchasing power, liquidity, recognizability, physical usability}). \quad (59)$$

This does not imply that biology determines monetary value. Rather, biological constraints help determine the human interface through which physical money is used.

14 Philosophy, Law, and Religion

Money raises philosophical questions concerning representation, trust, obligation, sovereignty, and the distinction between an object and a claim upon an object.

A coin is a physical object. A promissory note is also a physical object, but its monetary function may represent an obligation, a sovereign guarantee, or an institutionally recognized claim.

The distinction can be written schematically:

$$\text{object} \neq \text{claim},$$

even when both circulate as money.

Legal enforceability changes the state space of the system. Let $\mathcal{K}_t$ represent institutional credibility and legal capacity. Then

$$\mathcal{K}_t = \mathcal{K}(\text{law, enforcement, governance, institutional continuity}). \quad (60)$$

Religious traditions have historically considered questions concerning honesty, debt, usury, obligation, justice, trust, and stewardship. The present framework does not impose a single religious doctrine. Instead, such traditions may be treated as sources of ethical constraints and preference structures relevant to monetary institutions.

15 Political Economy and Sovereignty

Control over issuance can generate seigniorage.

Let issuance revenue be

$$S_t = I_t - c_C(I_t, t), \quad (61)$$

subject to expected redemption and financial-stability costs.

A political authority may therefore solve

$$\max_{\{I_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [S_t - \Phi(\chi_t) - \Omega_t], \quad (62)$$

where Ω_t represents macroeconomic and institutional costs.

The authority faces a time-consistency problem. A short-run incentive to increase issuance may conflict with long-run credibility.

16 International Relations and Geopolitical Monetary Architecture

Cross-border monetary systems introduce additional ratios.

Let country i hold promissory claims $C_{ij,t}$ supported directly or indirectly by the monetary base of country j .

Define

$$\chi_{ij,t} = \frac{C_{ij,t}}{M_{j,t}}. \quad (63)$$

The matrix

$$\mathbf{X}_t = [\chi_{ij,t}] \quad (64)$$

can be interpreted as a network of cross-jurisdictional monetary exposure.

Network stress can be represented using a weighted adjacency matrix $\mathbf{A}_t$:

$$\mathbf{s}_{t+1} = \mathbf{A}_t \mathbf{s}_t + \mathbf{b}_t + \mathbf{u}_t. \quad (65)$$

A domestic shock can therefore propagate internationally through trade, reserve holdings, convertibility arrangements, and financial confidence.

Diplomacy and institutional cooperation can alter the effective stability threshold by changing information, coordination, and reciprocal support.

17 Welfare Economics

Let social welfare be

$$\mathcal{W} = \mathcal{B}(\chi) - \Phi(\chi) - \mathcal{C} + \mathcal{A}, \quad (66)$$

where $\mathcal{B}$ denotes transaction benefits, Φ expected systemic losses, $\mathcal{C}$ production costs, and $\mathcal{A}$ accessibility and distributional benefits.

The planner solves

$$\max_{\chi, \mathbf{F}, \mathbf{M}} \mathcal{W}. \quad (67)$$

The optimality condition for χ is

$$\boxed{\frac{\partial \mathcal{B}}{\partial \chi} + \frac{\partial \mathcal{A}}{\partial \chi} = \frac{\partial \Phi}{\partial \chi} + \frac{\partial \mathcal{C}}{\partial \chi}}. \quad (68)$$

Thus the efficient degree of monetary layering need not coincide with the privately profitable degree of issuance.

18 Conflict, Warfare Economics, and Continuity Risk

The theory can be extended to extreme disruptions without assuming that monetary systems are designed primarily for warfare.

Let q_t denote a disruption state. Then

$$M_{t+1} = (1 - \delta_M(q_t))M_t + Q_t(q_t) - W_t(q_t), \quad (69)$$

while

$$C_{t+1} = (1 - \delta_C(q_t))C_t + I_t(q_t) - R_t(q_t). \quad (70)$$

Conflict can affect:

1. minting capacity;
2. printing capacity;
3. transport;
4. material supply;
5. institutional credibility;
6. communication networks;
7. redemption arrangements.

The resulting shock to χ_t can therefore arise from disruptions to either physical production or promissory issuance.

The relevant objective is continuity and resilience rather than monetary maximization alone.

19 Stage V: Econometric Identification

Define

$$x_t = \log C_t - \log M_t. \quad (71)$$

Suppose

$$x_t = a + \phi x_{t-1} + \beta_r r_t + \beta' \mathbf{Z}_t + u_t. \quad (72)$$

Define the identified ratio shock as

$$\boxed{\varepsilon_{\chi,t} = u_t}. \quad (73)$$

A local-projection model is

$$Y_{t+h} - Y_{t-1} = \alpha_h + \beta_h \varepsilon_{\chi,t} + \mathbf{\Gamma}'_h \mathbf{Z}_{t-1} + u_{t+h}. \quad (74)$$

The sequence

$$\{\beta_h\}_{h=0}^H$$

defines an impulse response.

Potential outcomes include:

$$\Delta \log Y_t, \quad (75)$$

$$\pi_t, \quad (76)$$

$$r_t, \quad (77)$$

$$\mathcal{S}_t, \quad (78)$$

$$\Delta \log C_t, \quad (79)$$

$$\Delta \log M_t. \quad (80)$$

A vector autoregression may be written

$$\mathbf{Y}_t = A_1 \mathbf{Y}_{t-1} + \cdots + A_p \mathbf{Y}_{t-p} + B \varepsilon_{\chi,t} + \mathbf{u}_t. \quad (81)$$

20 State-Space Representation

Let the latent state be

$$\mathbf{X}_t = (M_t, C_t, \chi_t, \Gamma_t, \delta_t, r_t, \sigma_t)'. \quad (82)$$

The transition equation is

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \mathbf{u}_{t+1}). \quad (83)$$

Observed variables satisfy

$$\mathbf{Y}_t = H(\mathbf{X}_t) + \mathbf{v}_t. \quad (84)$$

Linearization gives

$$\mathbf{X}_{t+1} = A\mathbf{X}_t + B\mathbf{u}_{t+1}, \quad (85)$$

$$\mathbf{Y}_t = C\mathbf{X}_t + D\mathbf{v}_t. \quad (86)$$

This representation permits Kalman filtering, nonlinear filtering, particle methods, Bayesian estimation, and machine-learning extensions.

21 Algorithms and Pseudo-Code

21.1 Cash-to-Coin Shock Estimation

INPUT:

```
Coin base series M[t]
Promissory-note series C[t]
Controls Z[t]
```

FOR each time t:

```
x[t] = log(C[t]) - log(M[t])
```

Estimate:

```
x[t] = a + phi*x[t-1] + beta'Z[t] + u[t]
```

FOR each time t:

```
epsilon_chi[t] = estimated residual u[t]
```

OUTPUT:

```
cash-to-coin shock series epsilon_chi[t]
```

21.2 Dynamic Stress Simulation

INITIALIZE:

```
M, C, Gamma, delta, r
```

FOR t = 1, ..., T:

```
chi = C / M
```

```
Gamma = confidence_transition(Gamma, chi, delta)
```

```
delta = redemption_function(Gamma, chi)
```

```
redemption = delta * C
```

```
M = (1 - attrition_coin)*M
    + coin_issuance
    - coin_withdrawal
```

```
C = (1 - attrition_cash)*C
```

```

+ note_issuance
- redemption

IF redemption > available_redemption_capacity:
    flag financial_stress

record all state variables

END FOR

```

22 Data Science and Machine Learning

The nonlinear state transition

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \mathbf{u}_{t+1})$$

need not be fully known.

A machine-learning approximation may be written

$$\widehat{\mathbf{X}}_{t+1} = \mathcal{N}_\theta(\mathbf{X}_t, \mathbf{Z}_t), \quad (87)$$

where $\mathcal{N}_\theta$ may represent a neural network.

However, unconstrained prediction may violate accounting identities. A hybrid architecture is therefore preferable:

$$\widehat{\mathbf{X}}_{t+1} = F_{\text{structural}}(\mathbf{X}_t) + \mathcal{N}_\theta(\mathbf{X}_t, \mathbf{Z}_t). \quad (88)$$

The accounting identity

$$\chi_t = \frac{C_t}{M_t} \quad (89)$$

is imposed structurally rather than learned approximately.

A probabilistic model may estimate

$$p(M_{t+1}, C_{t+1}, \Gamma_{t+1}, \delta_{t+1} \mid \mathbf{X}_t, \mathbf{Z}_t). \quad (90)$$

Anomaly detection can identify unusually large realizations of

$$|\varepsilon_{\chi,t}|.$$

23 A Reinforcement-Learning Extension

Suppose an issuing authority chooses actions

$$a_t = (I_t, Q_t, \mathbf{F}_t).$$

The state is $\mathbf{X}_t$ and the transition is

$$\mathbf{X}_{t+1} \sim P(\cdot \mid \mathbf{X}_t, a_t).$$

The authority maximizes

$$V^\pi(\mathbf{X}_t) = \mathbb{E}^\pi \left[\sum_{s=t}^{\infty} \beta^{s-t} R(\mathbf{X}_s, a_s) \right]. \quad (91)$$

The reward function may be

$$R = \text{liquidity utility} - \text{fragility cost} - \text{production cost} - \text{inflation cost}. \quad (92)$$

Safety constraints can impose

$$\Pr(\delta_t \chi_t > 1) \leq \alpha. \quad (93)$$

This converts the problem into constrained sequential decision-making.

24 A System Architecture

The complete system can be summarized graphically.

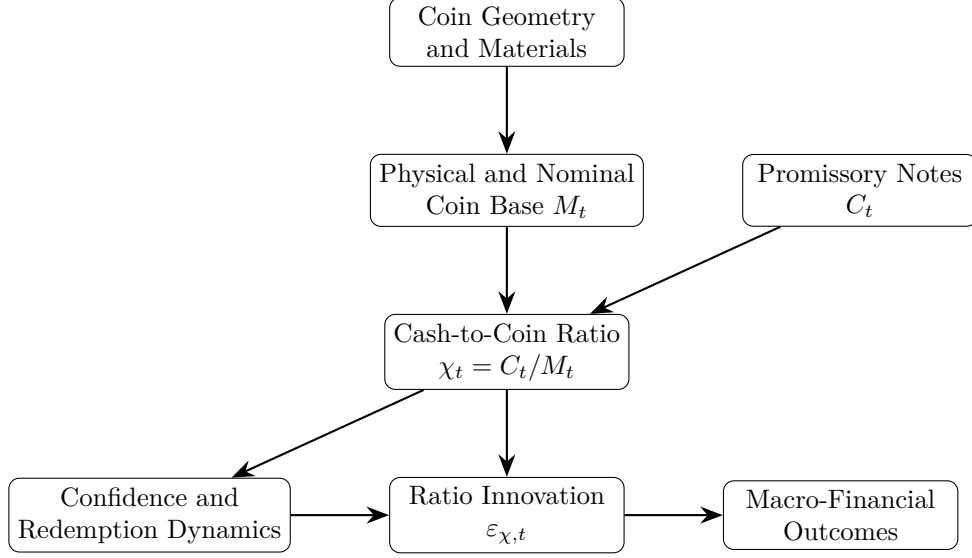


Figure 2: System architecture.

25 Integrated Financial Phase Space

The combined theory naturally defines a multidimensional phase space.

Let

$$\mathbf{Z}_t = (\hat{p}_t, r_t, \chi_t, \Gamma_t, \lambda_t, \mathbf{z}_t). \quad (94)$$

A regime map is

$$\mathcal{R}_t = \mathcal{R}(\mathbf{Z}_t). \quad (95)$$

The system can therefore undergo:

1. geometric transitions;
2. topological transitions;
3. backing-regime transitions;
4. confidence transitions;
5. redemption transitions;
6. macro-financial transitions.

The resulting theory is a coupled phase theory of physical and promissory money.

26 Main Theoretical Implications

The principal implications are as follows.

1. The ratio C_t/M_t is an economically meaningful state variable.
2. Unexpected changes in that ratio can be treated as monetary shocks.
3. A ratio shock has two mechanically distinct origins:

$$\varepsilon_{\chi,t} \approx \varepsilon_{C,t} - \varepsilon_{M,t}.$$

4. Higher χ increases monetary layering and transaction liquidity.
5. Higher χ reduces the fraction of simultaneous redemption that can be immediately satisfied by a fixed coin base.
6. Coin geometry can influence the denominator of the ratio through material requirements, physical stocks, production constraints, and installed-stock dynamics.
7. Negative interest rates need not have a universally predetermined effect on χ .
8. Confidence creates nonlinear feedback between monetary structure and redemption behavior.
9. Denomination composition matters even when aggregate cash-to-coin ratios are identical.
10. Monetary resilience depends jointly on physics, production, institutions, expectations, law, and macroeconomic policy.

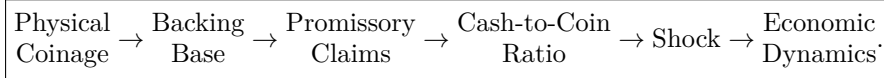
27 Conclusion

This paper has developed a theory of promissory money backed by coinage in which the central state variable is

$$\chi_t = \frac{C_t}{M_t}.$$

The central innovation is to interpret unexpected movements in this ratio as monetary and financial shocks.

The theory begins with a physical monetary base and adds a contractual layer. The resulting system has a natural architecture:



The framework connects geometry to monetary economics, manufacturing to financial stability, redemption mechanics to behavioral expectations, and physical monetary design to macroeconomic dynamics.

The most important general conclusion is that the distinction between money and a claim upon money is itself a source of dynamics. Once the relationship between the two is allowed to vary, the ratio between them becomes capable of generating amplification, fragility, regime changes, and empirically identifiable shocks.

The resulting research program is therefore not simply a theory of banknotes. It is a general theory of monetary layering.

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Glossary

Cash-to-coin ratio

The ratio

$$\chi_t = C_t/M_t$$

between outstanding promissory money and the relevant coin base.

Cash-to-coin shock

An unexpected innovation in $\log \chi_t$.

Coin base

The specified stock of coinage used as the denominator of the backing ratio.

Financial layering

The creation of promissory monetary claims on top of a physical monetary base.

Fractional backing

A regime in which promissory claims exceed the immediately available specified backing base.

Immediate redemption capacity

The maximum fraction of outstanding claims that can be simultaneously redeemed against the relevant base.

Monetary phase

A region of the state space characterized by qualitatively distinct backing, confidence, or redemption properties.

Nominal backing

Backing measured by face or legally specified monetary value.

Physical backing

Backing measured through the physical properties or mass of the underlying coin stock.

Market-value backing

Backing measured by the market value of underlying monetary assets.

Promissory money

A transferable monetary instrument representing an institutionally recognized promise, claim, or obligation.

Redemption pressure

The proportion δ_t of claims presented for redemption.

Spectral geometry

The representation of geometric structure through coefficients associated with spectral modes.

State variable

A variable required to characterize the evolving condition of a dynamical system.

The End